# Quiz 2 (Version 4)

CAS CS 132: *Geometric Algorithms*

October 5, 2026

Name:

BUID:

- ▷ You will have approximately 35 minutes to complete this quiz.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

# 1 Spans and Linear Equations

Determine a linearly independent set of vectors $V$ such that span $V$ is the same as the solution set of the following linear equation.

$$x - 5y + 2z = 0$$

*Solution.*

## 2 Dependence Relations

Let $A = [\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \mathbf{a}_3 \ \ \mathbf{a}_4]$ be a matrix (that is, $\mathbf{a}_1, \ldots, \mathbf{a}_4$ are the columns of $A$ from left to right) that is row equivalent to the matrix given below. Write a dependence relation for the columns of $A$. That is, write the vector $\mathbf{0}$ as a nontrivial linear combination of the columns of $A$.

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

*Solution.*

## 3 Matrix Equations

Determine a general form solution for the homogeneous equation $A\mathbf{x} = \mathbf{0}$ where $A$ is given below.

$$\begin{bmatrix} -2 & -2 & 2 \\ -3 & -2 & 4 \\ 2 & 1 & -3 \end{bmatrix}$$

*Solution.*

## 4 NumPy Array Slicing

Implement the function `quad` that, given a 2D NumPy array `A`, returns four 2D NumPy arrays corresponding to the four quadrants of `A`. For simplicity you may assume that `A` has an even number of rows and an even number of columns. In addition to array slicing you'll need to use the `shape` property and integer division (`//`). The following is an example of the expected behavior.

```
>>> A = np.array([[0, 1, 2, 3], [4, 5, 6, 7]])
>>> top_left, top_right, bottom_left, bottom_right = quad(A)
>>> top_left
array([[0, 1]])
>>> bottom_right
array([[6, 7]])
```

*Solution.*